

Report: **Bank Data Analysis**

Include your visualizations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

- 1.1.1. In order to read the HTML files from https://web.archive.org/web/20230908091635/https://en.wikipedia.org/wiki/List_of_largest_banks i used `pd.read_html`. There are 3 table in the wikipedia page. Following is the table 1

3 tables found

	Rank	Bank name	Market cap (US\$ billion)
0	1	JPMorgan Chase	432.92
1	2	Bank of America	231.52
2	3	Industrial and Commercial Bank of China	194.56
3	4	Agricultural Bank of China	160.68
4	5	HDFC Bank	157.91
5	6	Wells Fargo	155.87
6	7	HSBC Holdings PLC	148.90
7	8	Morgan Stanley	140.83
8	9	China Construction Bank	139.82
9	10	Bank of China	136.81

- 1.1.2. Shape of the data frame: (10,3)

- 1.1.3. Used `printSchema` to get details:

```

root
|-- Rank: long (nullable = true)
|-- Bank name: string (nullable = true)
|-- Market cap (US$ billion): double (nullable = true)

```

1.1.4. Renamed the columns as per assignment demand

Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)
1	JPMorgan Chase	432.92
2	Bank of America	231.52
3	Industrial and Co...	194.56
4	Agricultural Bank...	160.68
5	HDFC Bank	157.91
6	Wells Fargo	155.87
7	HSBC Holdings PLC	148.9
8	Morgan Stanley	140.83
9	China Constructio...	139.82
10	Bank of China	136.81

2. Data Cleaning

2.1. Handle Missing Values

2.1.1 Drop rows with missing values .

There are no missing values in the data. We do not need to drop rows with missing value.

Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)
0	0	0

2.2. Fixing Columns

2.2.1. Ensure all columns are properly named and formatted

Renamed columns as below

"Rank" to "Bank_Rank"

"Bank name" to "Bank_Name"

"Market cap (US\$ billion)" to "Market_Capitalisation (USD Billion)"

Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)
1	JPMorgan Chase	432.92
2	Bank of America	231.52
3	Industrial and Co...	194.56
4	Agricultural Bank...	160.68
5	HDFC Bank	157.91
6	Wells Fargo	155.87
7	HSBC Holdings PLC	148.9
8	Morgan Stanley	140.83
9	China Constructio...	139.82
10	Bank of China	136.81

2.3. Handling Outliers

2.3.1 Identify outliers in the dataset and explain why they are considered outliers. It is not necessary to remove them for this task, but mention your approach for handling them.

Detected Outliers:

Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)
1	JPMorgan Chase	432.92

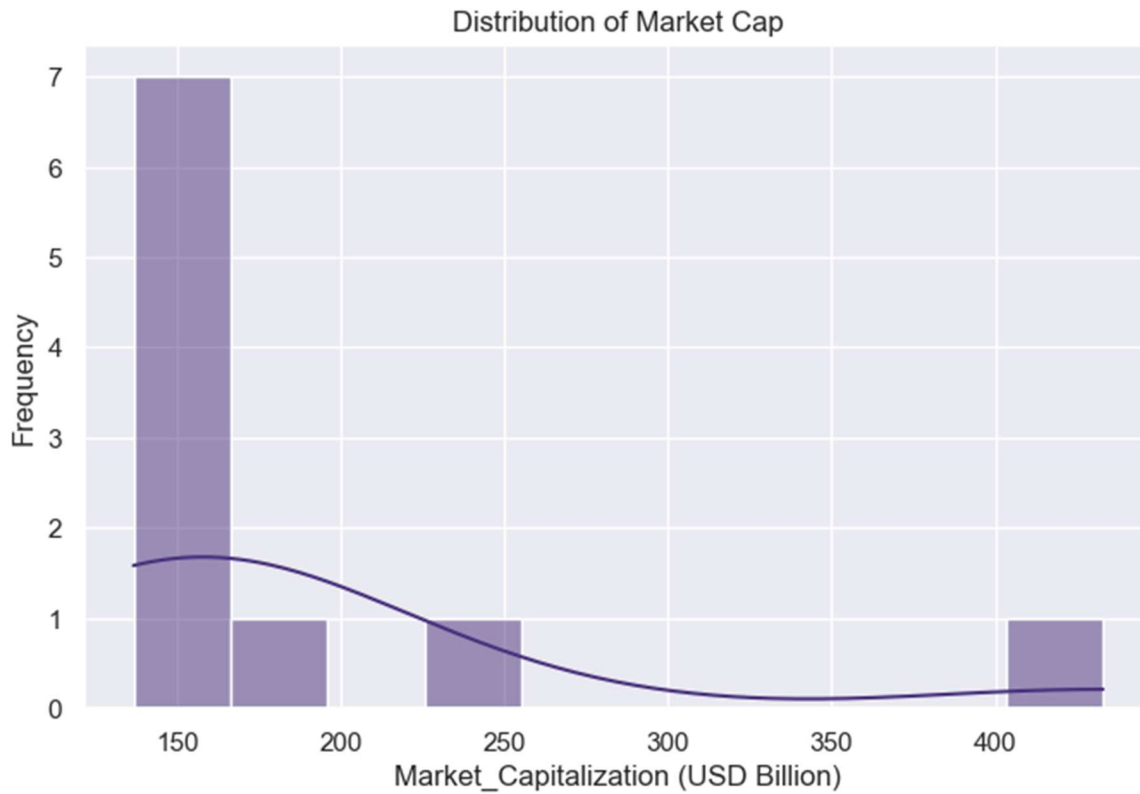
3. Exploratory Data Analysis

3.1. Convert PySpark DataFrame to Pandas DataFrame for Visualization.

	Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)
0	1	JPMorgan Chase	432.92
1	2	Bank of America	231.52
2	3	Industrial and Commercial Bank of China	194.56
3	4	Agricultural Bank of China	160.68
4	5	HDFC Bank	157.91

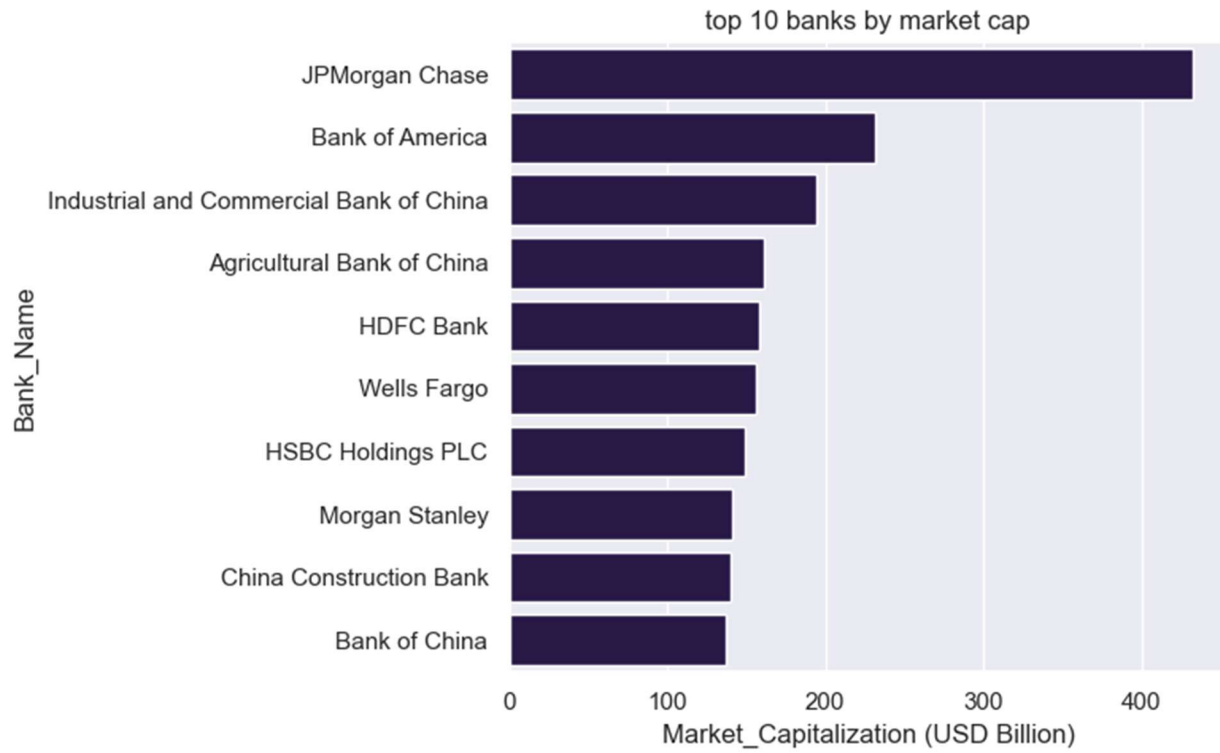
3.2. Analyze the distribution of market capitalization using a histogram.

```
Market Cap distribution
count      10.000000
mean       189.982000
std         90.223932
min         136.810000
25%         142.847500
50%         156.890000
75%         186.090000
max         432.920000
Name: Market_Capitalisation (USD Billion), dtype: float64
```



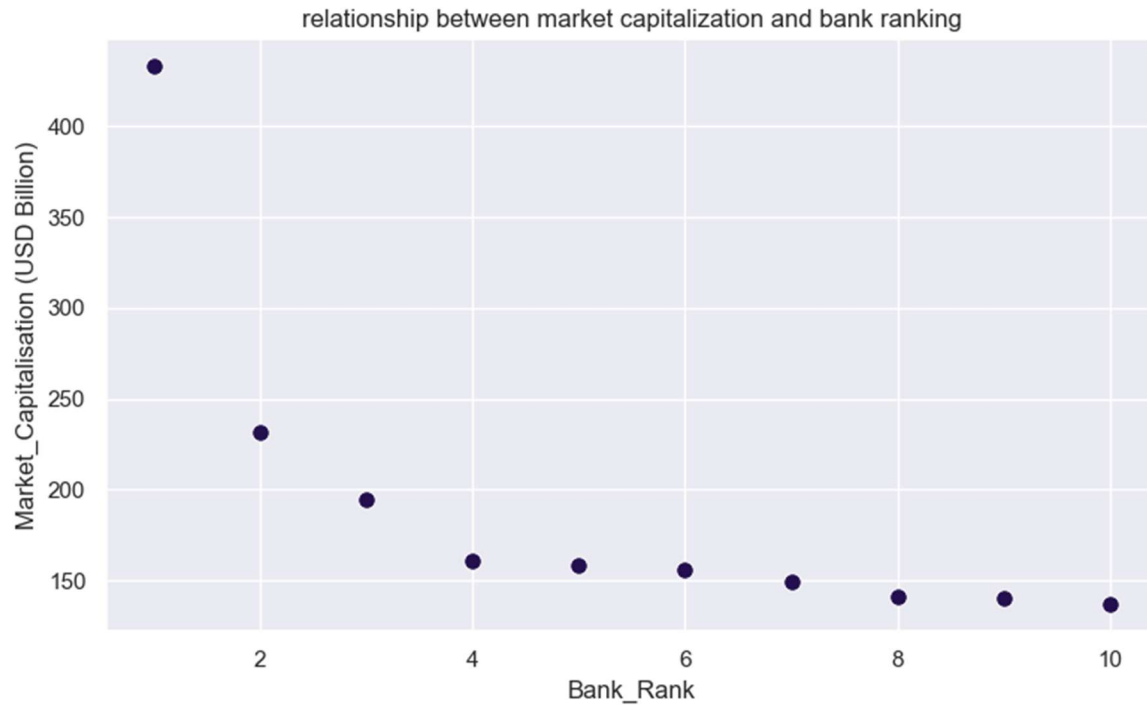
The histogram reveals **right-skewed distribution**, shows that a few banks (like JPMorgan Chase) have exceptionally high market capitalizations.

3.3. Identify the top 10 banks by market capitalization using a bar chart.



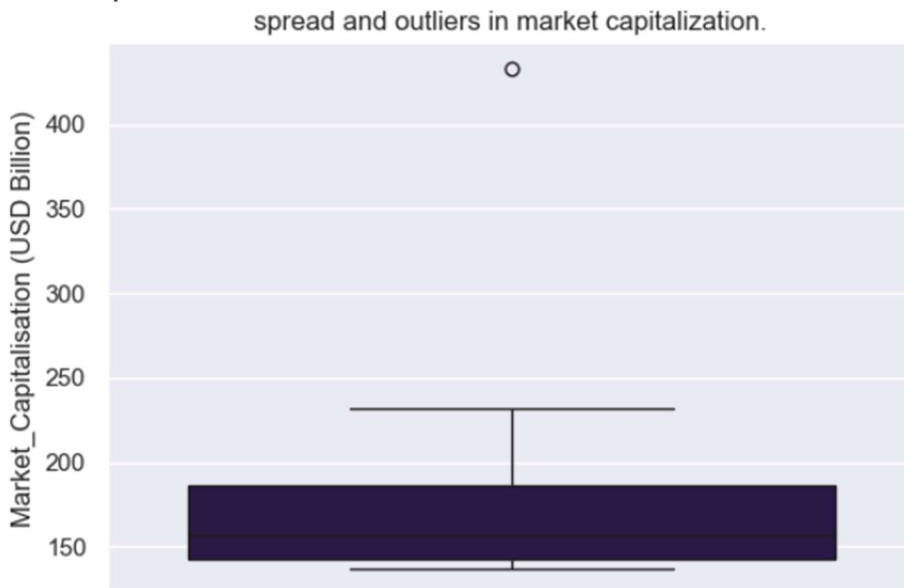
- JPMorgan Chase leads with a significant margin (~433 B USD), followed by Bank of America (~232 B USD).
- The gap between the first and second bank alone is over **200 B USD**, demonstrating a **high concentration of power** at the top.
- Chinese banks (ICBC, Agricultural Bank of China, Bank of China) collectively occupy multiple top-10 positions, underscoring China's strong global presence.

3.4. Visualize the relationship between market capitalization and bank ranking using a scatter plot.



- The scatter plot shows a clear **inverse relationship** — higher market capitalization corresponds to lower rank numbers (i.e., better rank).
- A steep drop-off after the top 3 indicates a **winner-takes-most** trend in global banking.

3.5. Use a boxplot to examine the spread and outliers in market capitalization.



➤ **Central Tendency (Median):**

- lies around **160–170 billion USD**, indicating that half of the banks have market capitalization below this level and half above.

➤ **Interquartile Range (IQR):**

- roughly **140 to 190 billion USD**.
- This means that **50% of all banks** have market capitalizations within this range.

➤ **Whiskers (Data Spread):**

- The upper whisker reaches approximately **250 billion USD**
- The lower whisker goes down to about **140 billion USD**, means the smallest observed non-outlier market cap.

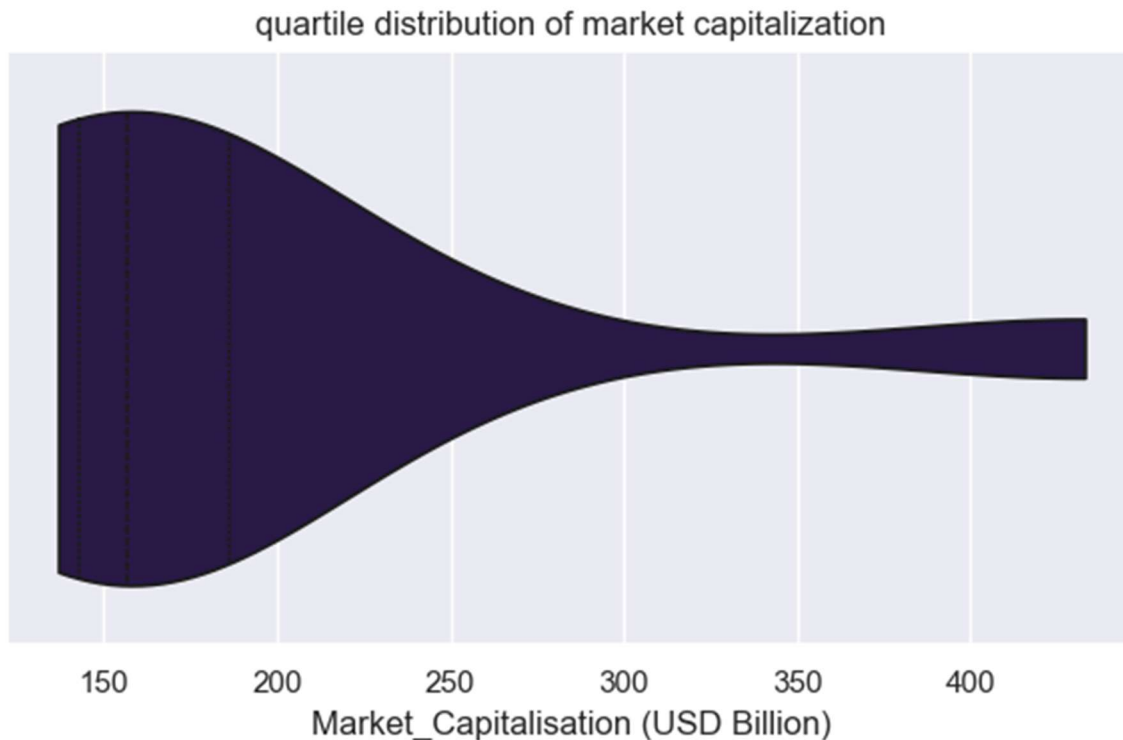
➤ **Outlier (Top Point):**

- The single **dot above the box (~430 billion USD)** represents an **outlier** — a value significantly higher than the rest.
- This corresponds to **JPMorgan Chase**

➤ **Overall Distribution:**

- Most banks cluster around mid-to-high hundreds (150–200 B USD), while only one or two fall in the “mega cap” category.

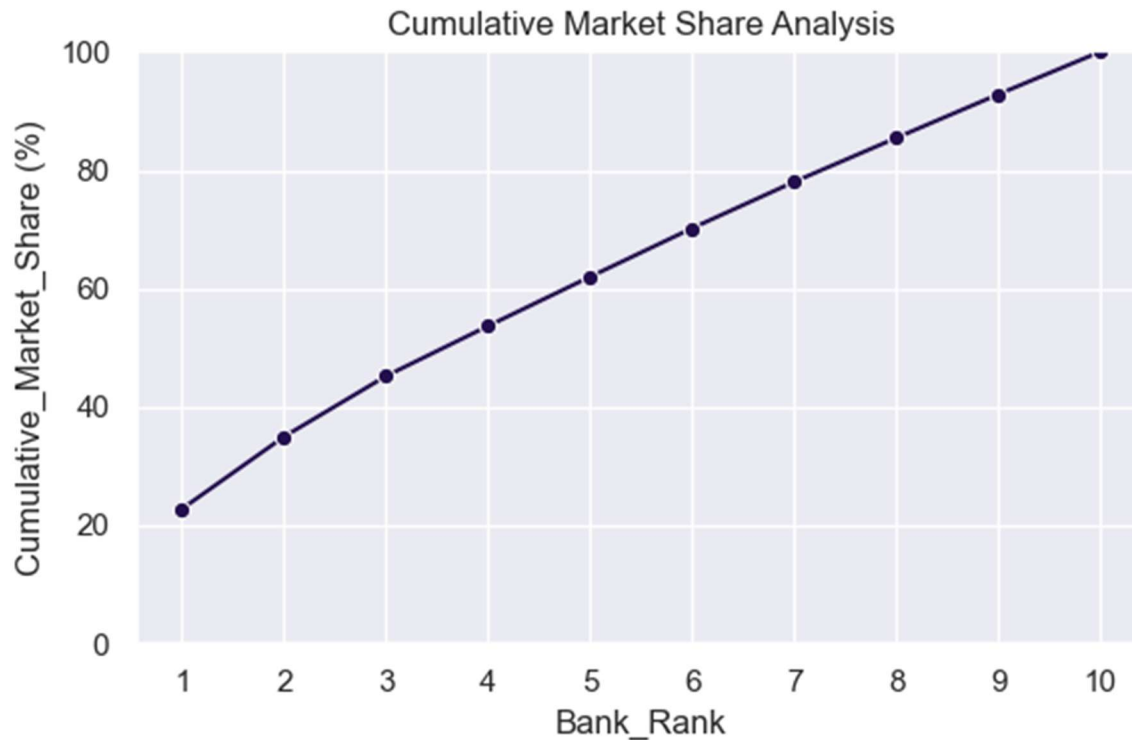
3.6. Display the quartile distribution of market capitalization using a violin plot.



The violin plot confirms that the **upper quartile** holds a disproportionate share of total market capitalization, indicating strong dominance by top players.

3.7. Compute cumulative market share and visualize it with a line plot.

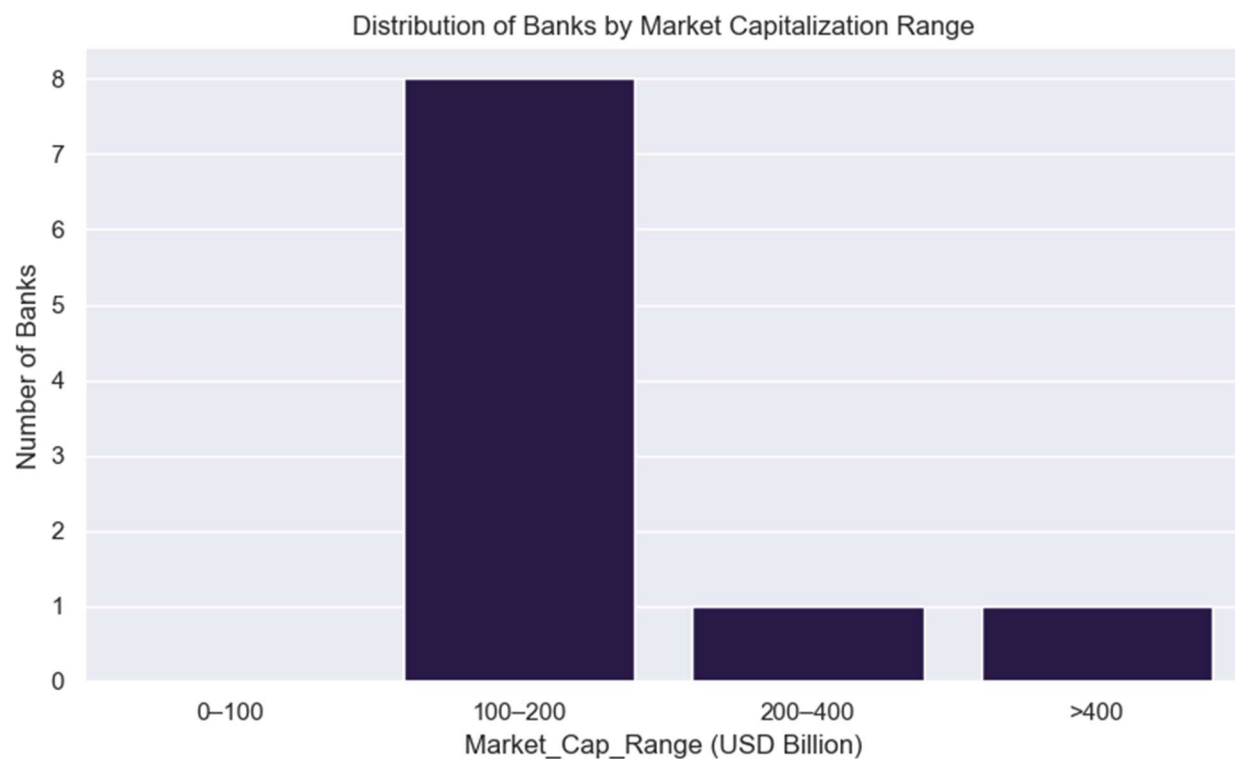
Bank_Rank		Bank_Name	Market_Capitalisation (USD Billion)	Cumulative_Market_Share
0	1	JPMorgan Chase	432.92	22.787422
1	2	Bank of America	231.52	34.973840
2	3	Industrial and Commercial Bank of China	194.56	45.214810
3	4	Agricultural Bank of China	160.68	53.672453
4	5	HDFC Bank	157.91	61.984293



- The scatter plot shows a clear **inverse relationship** — higher market capitalization corresponds to lower rank numbers (i.e., better rank).
- A steep drop-off after the top 3 indicates a **winner-takes-most** trend in global banking.

3.8. Categorize banks into market capitalization ranges and analyze their distribution using a bar chart.

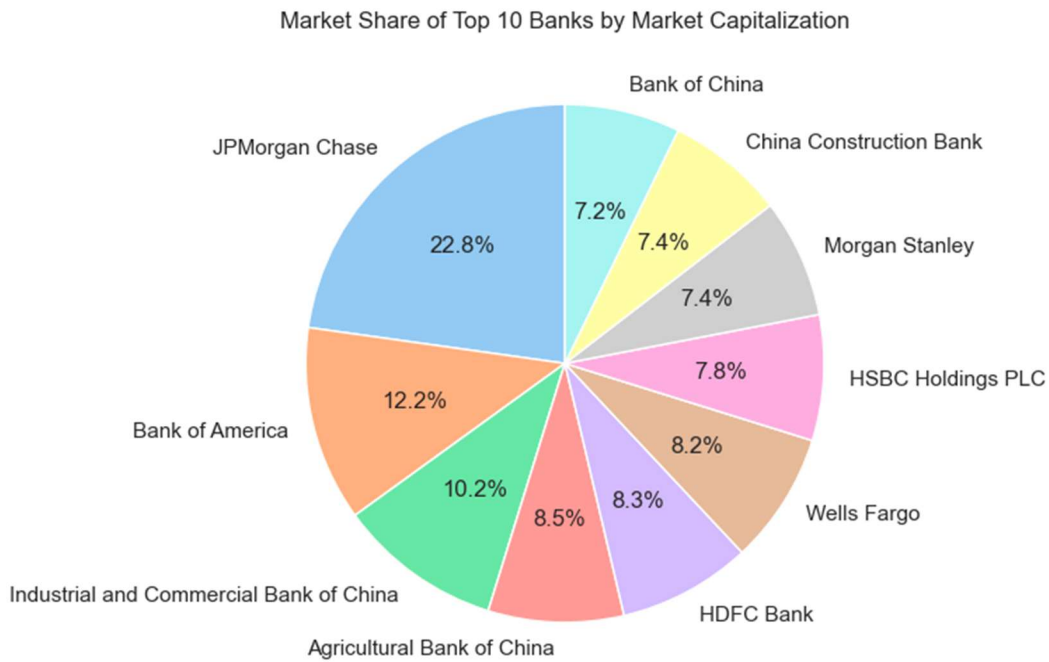
	Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)	Cumulative_Market_Share	Market_Cap_Range
0	1	JPMorgan Chase	432.92	22.787422	>400
1	2	Bank of America	231.52	34.973840	200–400
2	3	Industrial and Commercial Bank of China	194.56	45.214810	100–200
3	4	Agricultural Bank of China	160.68	53.672453	100–200
4	5	HDFC Bank	157.91	61.984293	100–200
5	6	Wells Fargo	155.87	70.188755	100–200
6	7	HSBC Holdings PLC	148.90	78.026339	100–200
7	8	Morgan Stanley	140.83	85.439147	100–200
8	9	China Construction Bank	139.82	92.798791	100–200
9	10	Bank of China	136.81	100.000000	100–200



- Majority of banks belong to the **Large Cap (100–300 B)** segment.
- Few banks qualify as **Mega Cap (>300 B)** or **Mid Cap (<100 B)**,

3.9. Calculate and display market share distribution of top 10 banks using a pie chart.

Bank_Rank		Bank_Name	Market_Capitalisation (USD Billion)	Market_Share_%
0	1	JPMorgan Chase	432.92	22.787422
1	2	Bank of America	231.52	12.186418
2	3	Industrial and Commercial Bank of China	194.56	10.240970
3	4	Agricultural Bank of China	160.68	8.457643
4	5	HDFC Bank	157.91	8.311840



4. Banking Data ETL Querying

4.1 Perform Advanced Market Capitalization Analysis with Growth Metrics.

Top Banks by Market Cap with Advanced Metrics:

Bank_Name	Market_Capitalisation (USD Billion)	Bank_Rank	market_share(%)	cumulative_market_cap	gap_from_leader
JPMorgan Chase	432.92	1	22.79	432.92	0.0
Bank of America	231.52	2	12.19	664.44	-201.4
Industrial and Co...	194.56	3	10.24	859.0	-238.36
Agricultural Bank...	160.68	4	8.46	1019.680000000001	-272.24
HDFC Bank	157.91	5	8.31	1177.590000000001	-275.01
Wells Fargo	155.87	6	8.2	1333.46	-277.05
HSBC Holdings PLC	148.9	7	7.84	1482.360000000001	-284.02
Morgan Stanley	140.83	8	7.41	1623.19	-292.090000000003
China Constructio...	139.82	9	7.36	1763.01	-293.1
Bank of China	136.81	10	7.2	1899.82	-296.11

4.2 Analyze Market Concentration and Categorize Banks Based on Market Share Tiers.

```

+-----+-----+-----+-----+
|Bank_Rank|      Bank_Name|Market_Capitalisation (USD Billion)|Market_Share(%)|
+-----+-----+-----+-----+
|      1|      JPMorgan Chase|      432.92|      22.79|
|      2|      Bank of America|      231.52|      12.19|
|      3|Industrial and Co...|      194.56|      10.24|
+-----+-----+-----+-----+
only showing top 3 rows

+-----+-----+
|Market_Share_Tier|count|
+-----+-----+
|      Tier-1: >5%|    10|
+-----+-----+

```

4.3 Examine Statistical Distribution of Market Capitalization Using Quartile Analysis.

Q1: 140.83, Median: 155.87, Q3: 194.56

```

+-----+-----+-----+-----+
|      Bank_Name|Market_Capitalisation (USD Billion)|Market_Cap_Quartile|
+-----+-----+-----+-----+
|      JPMorgan Chase|      432.92|      High (Q4)|
|      Bank of America|      231.52|      High (Q4)|
|Industrial and Co...|      194.56|      Upper-Mid (Q3)|
|Agricultural Bank...|      160.68|      Upper-Mid (Q3)|
|      HDFC Bank|      157.91|      Upper-Mid (Q3)|
|      Wells Fargo|      155.87|      Lower-Mid (Q2)|
|      HSBC Holdings PLC|      148.9|      Lower-Mid (Q2)|
|      Morgan Stanley|      140.83|      Low (Q1)|
|China Constructio...|      139.82|      Low (Q1)|
|      Bank of China|      136.81|      Low (Q1)|
+-----+-----+-----+-----+

```

4.4 Conduct Comparative Size Analysis to Classify Banks by Relative Market Size.

Bank_Name	Market_Capitalisation (USD Billion)	Market_Size_Category
JPMorgan Chase	432.92	Mega Cap
Bank of America	231.52	Large Cap
Industrial and Co...	194.56	Large Cap
Agricultural Bank...	160.68	Large Cap
HDFC Bank	157.91	Large Cap
Wells Fargo	155.87	Large Cap
HSBC Holdings PLC	148.9	Large Cap
Morgan Stanley	140.83	Large Cap
China Constructio...	139.82	Large Cap
Bank of China	136.81	Large Cap

Market_Size_Category	count
Mega Cap	1
Large Cap	9

Only JPMorgan qualifies as **Mega Cap**

4.5. Evaluate Market Growth and Identify Gaps Between Consecutive Banks.

Market Cap Gaps Between Consecutive Banks:

Bank_Name	Market_Capitalisation (USD Billion)	Bank_Rank	gap_to_next	gap_percentage
JPMorgan Chase	432.92	1	201.4	46.52
Bank of America	231.52	2	36.96	15.96
Industrial and Co...	194.56	3	33.88	17.41
Agricultural Bank...	160.68	4	2.77	1.72
HDFC Bank	157.91	5	2.04	1.29
Wells Fargo	155.87	6	6.97	4.47
HSBC Holdings PLC	148.9	7	8.07	5.42
Morgan Stanley	140.83	8	1.01	0.72
China Constructio...	139.82	9	3.01	2.15

4.6. Assess Market Dominance by Measuring Cumulative Share and Dominance Score.

Bank_Name	Market_Share(%)	Rank	Cumulative_Market_Share(%)	Dominance_Score	Dominance_Category
JPMorgan Chase	22.79	1	22.79	22.79	Dominant
Bank of America	12.19	2	34.98	6.095	Dominant
Industrial and Co...	10.24	3	45.22	3.4133333333333336	Major
Agricultural Bank...	8.46	4	53.68	2.115	Major
HDFC Bank	8.31	5	61.99	1.6620000000000001	Competitive
Wells Fargo	8.2	6	70.19	1.3666666666666665	Competitive
HSBC Holdings PLC	7.84	7	78.03	1.1199999999999999	Competitive
Morgan Stanley	7.41	8	85.44	0.92625	Niche
China Constructio...	7.36	9	92.8	0.8177777777777778	Niche
Bank of China	7.2	10	100.0	0.72	Niche

4.7. Analyze Segment-Wise Bank Performance Based on Market Capitalization Ranges.

Bank_Rank	Bank_Name	Market_Capitalisation (USD Billion)	Market_Share(%)	Market_Share_Tier	Market_Size_Category
1	JPMorgan Chase	432.92	22.79	Tier-1: >5%	Mega Cap
2	Bank of America	231.52	12.19	Tier-1: >5%	Large Cap
3	Industrial and Co...	194.56	10.24	Tier-1: >5%	Large Cap
4	Agricultural Bank...	160.68	8.46	Tier-1: >5%	Large Cap
5	HDFC Bank	157.91	8.31	Tier-1: >5%	Large Cap

only showing top 5 rows

Market_Size_Category	Number_of_Banks	Avg_Market_Cap	Total_Market_Cap	Total_Market_Share(%)
Large Cap	9	162.99	1466.9	77.21
Mega Cap	1	432.92	432.92	22.79

- Tier-1 banks (> 5 % share) dominate global banking; others hold minor individual shares.
- Dominance scores drop exponentially after the top 3 banks, highlighting a **steep hierarchy**.

4.8. Generate a Comprehensive Performance Dashboard for Bank Rankings and Metrics.

- The dashboard visualizes how **Market Size Category**, **Quartile**, and **Dominance Tier** align — Mega Cap banks also have the highest dominance and share tiers.
- Clear segmentation helps track which categories (e.g., Mid Caps) present **growth potential** for investors and regulators.
- Final dashboard dataset is saved in kaggle. (because aws was chargeable)
<https://www.kaggle.com/datasets/gayatribehera5474/cleaned-bank-data-dashboard>

Comprehensive Global Bank Market Dashboard



5. Conclusion:

Provide final insights and recommendations based on the analysis:

5.1 Recommendations to track and compare market capitalization of the top global banks to evaluate competitiveness and dominance.

The analysis shows that a few global banks, especially **JPMorgan Chase**, **Bank of America**, and **ICBC**, dominate the global banking market with very high market capitalizations. To evaluate competitiveness, banks should track their market cap and ranking on a regular basis. Comparing quarterly or yearly data will help identify changes in performance and detect new emerging leaders. This continuous tracking will also help banks and investors understand shifts in dominance within the sector.

5.2 Suggestions to use cross-currency analysis (USD, GBP, EUR, INR) for consistent benchmarking of financial institutions across regions.

Since banks operate in different countries and currencies, comparing their values in one currency (like USD) may not always give the correct picture. A cross-currency analysis using **USD, GBP, EUR, and INR** ensures fair and consistent comparison across regions. It also helps evaluate the real value of banks when exchange rates fluctuate. This approach makes benchmarking more accurate and globally relevant.

5.3 Propose continuous monitoring of market share concentration to identify growth opportunities for mid-tier banks.

The market share analysis indicates that a small number of top banks hold a very large portion of the total global market capitalization, while mid-tier banks control a much smaller share. To improve competitiveness, it is important to monitor market share concentration continuously.

5.4 Identify potential regions or banking segments for expansion by analyzing gaps between tiers of banks and regional trends.

From the quartile and market size analysis, it is clear that most banks belong to the large-cap category, while only one or two reach the mega-cap level. Regions like **Asia-Pacific and India** show strong potential for future growth. Mid-sized banks can focus on these markets, digital transformation, and cross-border collaboration to strengthen their position and move into higher capitalization tiers.